UNIVERSITY OF FLORIDA
Mechanical and Aerospace Engineering

EML 4930/6934 Sensor-based Robot Planning
Exam 2 (Take-Home)

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Assigned: Wednesday, March 11th, 2026
Due: 11:59pm on Monday, March 23th, 2026

This exam is open notes and open internet (e.g., Google), and you may use MATLAB or Python. The use of generative AI tools is permitted; however, any risks associated with their use—including incorrect, misleading, or fabricated information—are the sole responsibility of the student. You are expected to verify the accuracy of any AI-generated content before including it in your answers. All sources, including generative AI tools, must be properly cited. Any unauthorized assistance or failure to provide appropriate citations will be considered plagiarism and handled in accordance with university academic integrity policies.

Show all your work. You will not get any credit for illegible or incomplete writing. A correct answer without any work will not receive any credit. Full credit will be given only if all steps required to reach a solution are shown.

You are not allowed to discuss this exam with anyone else, whether they are students in the course or not.

This exam is comprised of two parts. In Part I, you may not use MATLAB or Python. In Part II, you are required to use MATLAB or Python to implement path planning algorithms and answer the questions.

Your solutions and codes must be submitted through Canvas. If you have any questions regarding this exam, you must post your questions through the designated Canvas Discussion page.

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NAME (Please Print):

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Sign the UF Honor Pledge, "On my honor, I have neither given nor received unauthorized aid in doing this examination.

1 Part I (40 pts in total)

1.1 Bayes Filter (10 pts)

You are given a robot that is equipped with a vacuum unit to clean the floor. The robot has a binary sensor that detects whether the floor tile is clean or dirty. However, neither the cleaning unit nor the sensor are perfect. From previous experience, you know that the robot succeeds in cleaning a dirty floor tile with a probability of

$$P(x_{t+1} = clean | x_t = dirty) = 0.6 \tag{1}$$

where x_t is the state of the floor tile at time t and x_{t+1} is the resulting state after the action has been applied.

Activating the cleaning unit when the tile is clean will never make it dirty. Assume the robot always cleans at every time t (i.e. the transition probabilities model the fact that the robot is cleaning the floor tile). The probability that the sensor indicates that the floor tile is clean although it is dirty is $p(z_t = clean | x_t = dirty) = 0.2$ and the probability that the sensor correctly detects a clean tile is $p(z_t = clean | x_t = clean) = 0.6$.

Unfortunately, you have no knowledge about the current state of the floor tile. However, after cleaning the tile, the robot's sensor indicates that it is clean. Compute $p(x_{t+1} = clean | z_{t+1} = clean)$, i.e., the probability that at time $t+1$, the floor tile is now clean given that the sensor indicates it is clean. Assume a prior distribution at time t as $p(x_t = clean) = c$, where $0 \leq c \leq 1$. Then, using a software package of your choice, plot $p(x_{t+1} = clean | z_{t+1} = clean)$ for $0 \leq c \leq 1$. You do not need to submit your code for plotting.

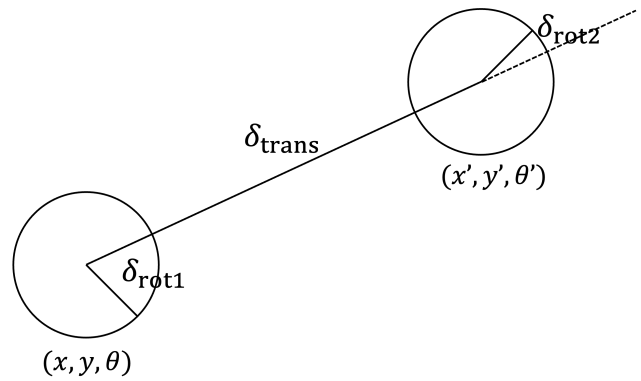

Figure 1: Robot model given in Lecture 17

1.2 Extended Kalman Filter (30 pts)

We will use the robot motion model introduced in Lecture 17 (Figure 1). The state of the robot is its 2D position and orientation: $s = [x, y, \theta]^T$. The control to the robot is $u = [\delta_{rot1}, \delta_{trans}, \delta_{rot2}]$, i.e. the robot rotates by δ_{rot1}, drives straight forward δ_{trans}, then rotates again by δ_{rot2}.

The equations for the motion model $s_t = g(u_t, s_{t-1})$ are as follows:

$$x_t = x_{t-1} + \delta_{trans} \cos(\theta_{t-1} + \delta_{rot1}) \tag{2}$$
$$y_t = y_{t-1} + \delta_{trans} \sin(\theta_{t-1} + \delta_{rot1}) \tag{3}$$
$$\theta_t = \theta_{t-1} + \delta_{rot1} + \delta_{rot2} \tag{4}$$

Derive the Jacobian matrix G, a matrix consisting of the first-order derivative of s_t with respect to the previous state s_{t-1}, as well as V, a matrix consisting of the first-order derivative of s_t with respect to the control input u_t. In short, derive the Jacobians of g with respect to the state $G = \frac{\partial g}{\partial s}$ and control $V = \frac{\partial g}{\partial u}$:

$$G = \begin{bmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} & \frac{\partial x'}{\partial \theta} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} & \frac{\partial y'}{\partial \theta} \\ \frac{\partial \theta'}{\partial x} & \frac{\partial \theta'}{\partial y} & \frac{\partial \theta'}{\partial \theta} \end{bmatrix}, \; V = \begin{bmatrix} \frac{\partial x'}{\partial \delta_{rot1}} & \frac{\partial x'}{\partial \delta_{trans}} & \frac{\partial x'}{\partial \delta_{rot2}} \\ \frac{\partial y'}{\partial \delta_{rot1}} & \frac{\partial y'}{\partial \delta_{trans}} & \frac{\partial y'}{\partial \delta_{rot2}} \\ \frac{\partial \theta'}{\partial \delta_{rot1}} & \frac{\partial \theta'}{\partial \delta_{trans}} & \frac{\partial \theta'}{\partial \delta_{rot2}} \end{bmatrix} \tag{5}$$

Then, assume there is a landmark m at location (l_x, l_y). The robot receives one measurement of the bearing angle ϕ , where

$$\phi = \tan^{-1}(\frac{l_y - y_t}{l_x - x_t}) - \theta_t \tag{6}$$

Derive Jacobian matrix H, derivative of the measurements with respect to the robot state $s_t = [x_t, y_t, \theta_t]$.

2 Part II (60 pts in total)

In Part II, you will implement extended Kalman filter and particle filters based on the work developed in Part I. You are given a code skeleton that you can fill in your own implementation of Kalman filter and particle filter. Fill in the Jacobian matrices G, V and H in the provided code using your derivation in Problem 1.2.

2.1 Extended Kalman Filter Implementation

2.1(a) Modeling EKF: Write down the pseudo code that you will implement.

2.1(b) Implementation: Plot the real robot path and the filter path under the default (provided) parameters.

2.2 (EML6934 ONLY) Particle Filter Implementation

2.2(a) Pseudo code: Write down the pseudo code that you will implement.

2.2(b) Implementation: Plot the real robot path and the filter path under the default (provided) parameters.